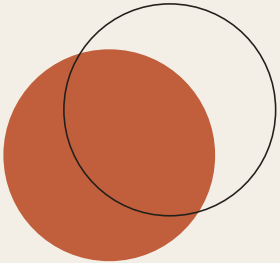
November 9, 2026

Research with AI Agents

What Works, What It Costs, and What an Institute Needs

Lukas Theissinger



Outline

I. OpenAI and Agent Usage

II. Owning Intelligence and Infrastructure at Lamarr

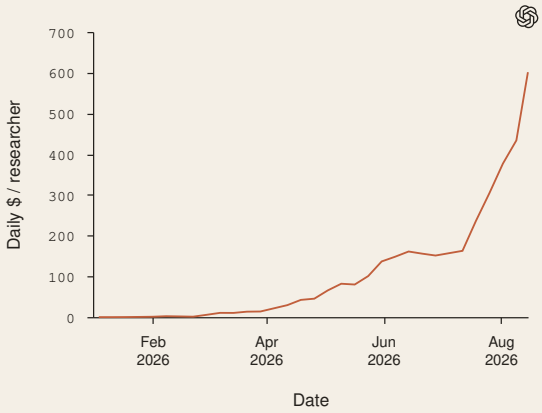


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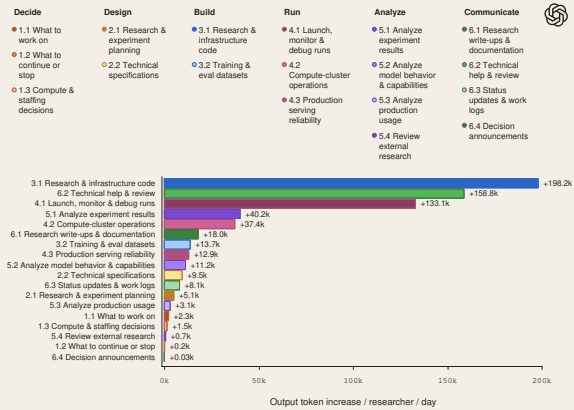
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FOMO Headline: OpenAI Agent Usage



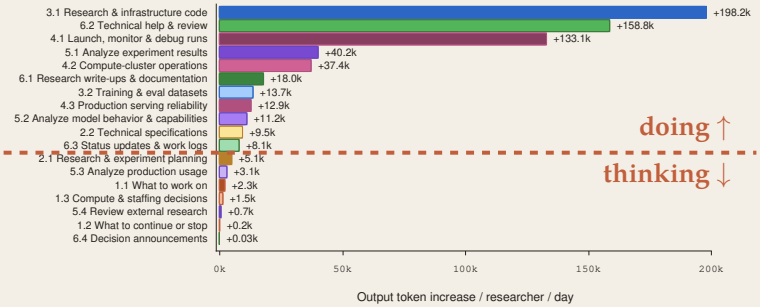
Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported)

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, “Research acceleration” (self-reported)

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What the OpenAI chart suggests

All 18 activities went up: agents are used everywhere, but how much and how exactly is still open

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What the OpenAI chart suggests	Rules of thumb
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open	Use agents or be left behind
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Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure	Agent use is a *skill*: not everything can or should be automated (degree of automation)

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Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
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4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it

II

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Automating at Bigger Scale

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Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** × **3 h per person per year**. An agent that reads the receipts and files the form correctly needs ∼**200 h** (rules, forms, access to the admin system)

$$\frac{\text{how often} \times \text{time saved}}{\text{time to build}}$$

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Lamarr, ∼300 researchers	$300 \times 9\,\text{h} = 2\,700\,\text{h/year}$	200 h	**pays >13×**

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So let's talk about costs

Decision 1: Subscription or API?

Subscription (20–200 €/month)

- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
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One month of my usage, priced at API list prices

	subscription	same usage via API	factor
Claude	20 $ (Pro)	∼680 $	∼34×
Codex	20 $ (Plus)	∼145 $	∼7×

Decision 2: Self-Hosting?

In the news

> “Revolut turns to self-hosted AI to cut coding costs tenfold over Claude”

Computing (UK), 2026

Decision 2: Self-Hosting? A Field Report: Pros and Cons

Hardware	Model	Quality	Speed
4 × H100	**GLM 5.3 Flash**	**Sonnet high**	**50–100 tokens/s**
repurposed	mainly	how it feels	per user

What I would use it for

- Pipelines, e.g. **AutoExercises**, finding related work
- High token, lower intelligence: Code implementation, Organizing experiments
- **Private Data**
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What I would not use it for

- High-intelligence, low-token work: Math proofs, idea finding
- Remember: OpenAI usage diagram

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To get a feeling for the headline: energy and the Sonnet comparison are **true** self-hosting values.
We do not go into all the details of self-hosting.
Repurposing hardware also takes skill and neat tricks.

GLM-5.3-Flash (main model)	**per month**
energy (full load)	∼330–670 €
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Payback: 4 × H100 at ∼150 k€ pays for itself in ∼10–22 months

Takeaways

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3. The best setup is **personal and custom**, so build your own